

Linux Hardening Audit Tool - Week 3 Update Report

Objective

Enhance the audit tool with advanced usability features, structured execution control, and compliance-oriented improvements.

Enhancements Implemented

- Command-Line Interface (CLI) support
- Quick and Full Scan Modes
- Selective Report Export Options
- CIS Benchmark Mapping for security checks
- Improved output formatting

Command-Line Interface

The tool now supports command-line arguments using argparse, allowing flexible execution modes and making it behave like professional security tools.

Scan Modes

Two scanning modes were introduced:

- Quick Scan: Performs essential checks for rapid assessment
- Full Scan: Executes all security checks for comprehensive auditing

Selective Report Export

Users can now export reports in JSON, HTML, or both formats using CLI options, improving usability and flexibility.

CIS Benchmark Mapping

Each security check is now associated with relevant CIS benchmark controls, enhancing compliance visibility and aligning the tool with industry standards.

Output Improvements

The audit output has been structured into clear sections with improved readability, making it easier to interpret results.

Impact

These enhancements transformed the script into a structured security auditing tool with flexible execution and compliance awareness.

Conclusion

Week 3 updates significantly improved the usability, flexibility, and professional quality of the audit tool, aligning it with real-world security auditing practices.